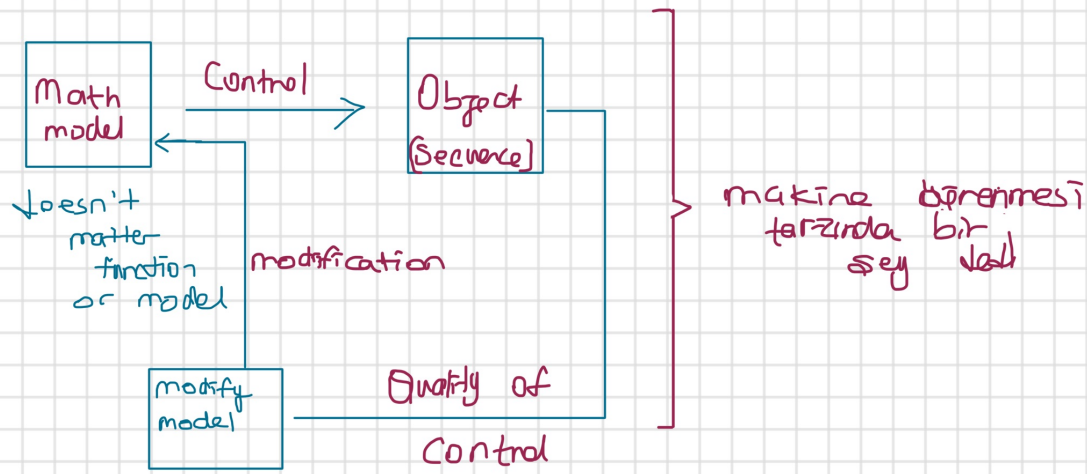


Week 3:



Teorik kısmı:

machine learning:

Supervised learning:

Special set

System try to find pattern.

* Classification

Reinforcement learning:

This give specific

Self-learning

for the next lesson there will be more theoretical presentation, and will be asked questions

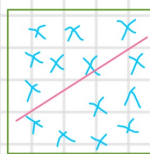
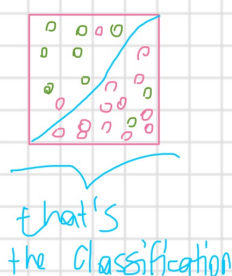
Supervised learning:

In supervised learning, the data analyst works with a collection of labeled examples $\{(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)\}$.

Each element x_i among N is called a feature vector.

The prediction of a class is classification but finding or predicting a real number is a regression.

The difference is that: In classification the learning algorithm looks for a line that separates examples of different classes from one another. In regression, on the other hand, the learning algorithm looks to find a line or a hyper-surface that closely follows the training examples.



that's the regression

Unsupervised learning

In unsupervised learning, the dataset is a collection of unlabeled examples $\{x_1, x_2, \dots, x_N\}$.

Again, x is a feature vector, and the goal of an unsupervised learning algorithm is to create a model that takes a feature vector x as input and either transforms it into another vector or into a value that can be used to solve a practical problem.

Semi-Supervised Learning:

In semi-supervised learning, the dataset contains both labeled and unlabeled examples. Usually, the quantity of unlabeled examples is much higher than the number of labeled examples. The goal of a semi-supervised learning algorithm is the same as the goal of the supervised learning algorithm.

Reinforcement Learning:

Reinforcement learning is a subfield of machine learning where the machine (called an agent) "lives" in an environment and is capable of perceiving the state of that environment as a vector of features. The machine can execute actions in non-terminal states. Different actions bring different rewards and could also move the machine to another state of the environment. A common goal of a reinforcement learning algorithm is to learn an optimal policy.

Self learning:

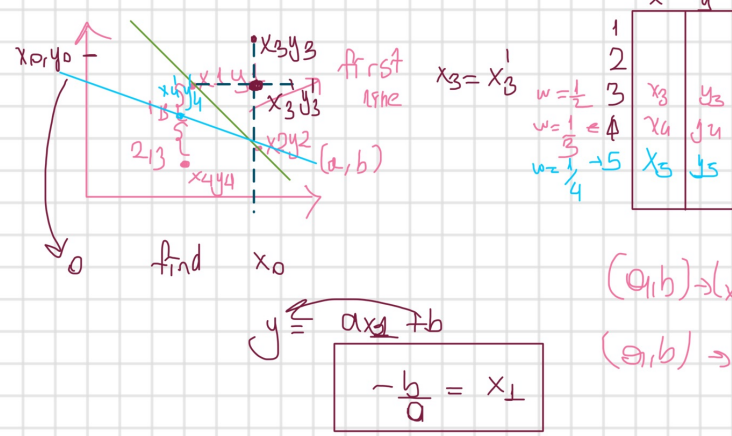
In a nutshell, self-learning models are AI models that, once deployed, can be optimised by training them on data that becomes more available over time.

Raw data:

Raw data is a collection of entities in their natural form; they cannot always be directly employable for machine learning.

Tidy data:

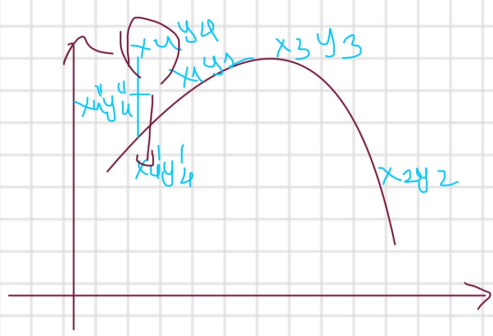
To be employable in machine learning, a necessary (but not sufficient) condition for the data is to be tidy. Tidy data can be seen as a spreadsheet, in which each row represents one example, and columns represent various attributes of an example.



$(0, b) \rightarrow (x_0, y_0, x_3''', y_3''')$
 $(a, b) \rightarrow (x_0, y_0, x_2, y_2)$

$\frac{1}{10} \rightarrow$ 11. adımdan sonra hep $\frac{1}{10}$ olarak gidecek

$(x_3'' y_3'' ; x_2 y_2) \rightarrow a, b$
 $(x_4'' y_4'' ; x_3'' y_3'') \rightarrow a, b$
 $\vdots$
 $(x_{30} y_{30} ; x_2 y_2)$



$$\begin{bmatrix} x_1^2 & x_1 & 1 \\ x_2^2 & x_2 & 1 \\ x_3^2 & x_3 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$A \quad \times \quad B$

$$x = \text{inv}(A) * B$$

We will give more feature our model to find now it act.

$$\begin{bmatrix} x_1^2 & x_1 & 1 \\ x_2^2 & x_2 & 1 \\ x_3^2 & x_3 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$A \quad \times \quad B$

$$\text{inv}(A) * B = x$$

Classification

$\left. \begin{matrix} 3, 2, 1 \\ 4, 3, 2 \\ 5, 4, 3 \end{matrix} \right\} 30 \text{ line}$